

Notes: Maenhout, Xing, and Balter (JF, 2026)

Model Ambiguity versus Model Misspecification in Dynamic Portfolio Choice

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1 Big picture

- **Research question.** How do investors who fear model uncertainty choose dynamic portfolios? More specifically, what changes when investors worry about *model misspecification* rather than only *model ambiguity*?
- **Main distinction.** Model ambiguity means the investor considers adverse alternatives inside a structured set of models. Model misspecification means the investor also allows broader, unstructured alternatives that can violate the benchmark model's dynamic restrictions.
- **Baseline intuition.** Even if benchmark returns are IID, a misspecification-averse investor may fear that returns are not IID. Nature chooses the worst-case dynamics, not just the worst-case mean.
- **Main theoretical finding.** Risk-averse investors with relative risk aversion $\gamma > 1$ endogenously fear return persistence, because persistence removes the possibility of intertemporal hedging. Risk-tolerant investors with $0 < \gamma < 1$ fear mean reversion, because mean reversion removes the possibility of intertemporal speculation.
- **Log utility result.** A log investor is myopic. Because the log investor does not care about intertemporal hedging or speculation, model misspecification collapses to model ambiguity.
- **Key state variable.** The paper uses *continuation entropy* as an endogenous state variable. This allows Nature to allocate model distortion over time and across states while preserving homotheticity.
- **Quantitative implications.** Model misspecification reduces optimal risky investment and consumption more than model ambiguity. It can generate belief scarring, endogenous nonparticipation in equity markets, and extrapolative return expectations.
- **Economic takeaway.** Robust portfolio choice depends not only on how pessimistic investors are about current expected returns, but also on which future return dynamics they fear.

2 Incremental contribution of the paper

- **Separating ambiguity from misspecification.** The paper makes precise the difference between structured model ambiguity and broader model misspecification in dynamic portfolio choice.
- **Constraint approach with homotheticity.** Instead of relying on the standard fixed-penalty multiplier approach, the paper uses entropy constraints. This preserves scale invariance and makes CRRA portfolio choice tractable beyond log utility.
- **Continuation entropy as a state variable.** The paper adapts the continuation-entropy idea from Hansen et al. to a homothetic portfolio setting. This is the central

technical device that makes model misspecification recursive and tractable.

- **Why log utility is not enough.** Much of the robustness literature uses log utility for tractability. The paper shows that this choice removes the economic force that distinguishes model misspecification from model ambiguity.
- **Endogenous belief dynamics.** The model generates extrapolative beliefs for risk-averse investors and contrarian beliefs for risk-tolerant investors, even when the benchmark model has IID returns.
- **Connection to existing robust portfolio rules.** The paper shows that homothetic robustness in Maenhout-style models corresponds to model ambiguity, not full model misspecification.
- **Applications.** The framework explains dynamic belief scarring, nonparticipation after bad return experiences, and reduced market timing when investment opportunities are stochastic.
- **Extension beyond IID returns.** The paper extends the analysis to a mean-reverting Sharpe ratio, where misspecification concerns remain tractable but ambiguity requires more restrictive rectangularity assumptions.

3 Model objects and measurement

3.1 Benchmark model, alternative models, and entropy

The paper starts from a benchmark probability model B . The investor worries that this model may not describe the true return-generating process. Alternative models U are measured by relative entropy, which penalizes deviations from the benchmark.

In continuous time, an alternative model is generated by a belief-distortion process u_t :

$$\frac{dU}{dB}\Big|_{\mathcal{F}_T} = Z_T = \exp\left(-\int_0^T \frac{1}{2}u_t^2 dt - \int_0^T u_t dB_t^B\right).$$

- $u_t > 0$ lowers the subjective expected return of the risky asset under the alternative model.
- Larger $|u_t|$ means the alternative model is further from the benchmark.
- Entropy constrains how far Nature can move from the benchmark.

Interpretation. Entropy is the model-uncertainty budget. The main question is whether Nature can spend that budget in a state-dependent way.

3.2 Model ambiguity versus model misspecification

- **Model ambiguity.** The investor considers structured alternatives. In the IID binomial tree, this means changing the up probability but keeping the IID structure.

- **Model misspecification.** The investor considers unstructured alternatives. In the IID binomial tree, this allows transition probabilities to depend on the previous return state.
- **Rectangularity.** Model ambiguity is connected to rectangular sets of priors, which impose pathwise restrictions and prevent Nature from reallocating entropy across states.
- **Key economic difference.** Misspecification lets Nature distort not just today's expected return but the dynamics of future expected returns.

Interpretation. Ambiguity is about which model inside a structured class is right. Misspecification is about whether the structured class itself is wrong.

3.3 Investor preferences

The investor has CRRA utility:

$$U(W) = \frac{W^{1-\gamma}}{1-\gamma}, \quad \gamma \neq 1,$$

with log utility as the limiting case when $\gamma = 1$.

- $\gamma > 1$: the investor is risk-averse and wants to hedge intertemporally.
- $0 < \gamma < 1$: the investor is risk-tolerant and likes intertemporal speculation.
- $\gamma = 1$: the investor is log and behaves myopically.

Interpretation. The sign of the investor's nonmyopic demand determines which return dynamics are most damaging.

3.4 Continuation entropy

Continuation entropy is defined as

$$E_t = \mathbb{E}_t^U \left[\int_t^\infty \delta e^{-\delta(s-t)} \frac{1}{2} u_s^2 ds \right].$$

It follows the dynamics

$$dE_t = \delta E_t dt - \frac{\delta}{2} u_t^2 dt + g_t dB_t^U.$$

- E_t is the remaining model-distortion budget from date t onward.
- u_t controls the instantaneous belief distortion.
- g_t controls how continuation entropy moves with return shocks.
- Allowing Nature to choose g_t is what creates model misspecification effects.

Interpretation. If Nature can choose g_t , then pessimism can become state-dependent. If $g_t = 0$, the model behaves like ambiguity with a fixed structured uncertainty set.

3.5 Quantitative environments

The paper studies three main environments.

- **Two-period binomial tree.** This gives transparent intuition for how entropy constraints differ between ambiguity and misspecification.
- **Continuous-time Merton model.** The risky asset has constant expected return and volatility. This shows how misspecification changes optimal investment and consumption.
- **Mean-reverting Sharpe ratio.** The investment opportunity set varies over time. This extension shows how misspecification affects market timing and exit.

Interpretation. The binomial model explains the mechanism; the continuous-time models show its quantitative and dynamic implications.

4 Methodology and theoretical specifications

4.1 Two-period binomial tree

Under the baseline model, returns are IID. A high return H occurs with probability p , and a low return L occurs with probability $1 - p$. The risk-free gross return satisfies $H > R_f > L$.

- Under model ambiguity, the investor is unsure about p , but the same distorted probability applies at all nodes.
- Under model misspecification, Nature can choose q_0 , q_H , and q_L . The probability of a high return in the second period can depend on whether the first-period return was high or low.

Interpretation. Misspecification allows the worst-case model to create serial dependence even when the benchmark model is IID.

4.2 Single entropy constraint: model misspecification

Model misspecification is represented by a single intertemporal entropy constraint. In the binomial tree, this lets Nature allocate entropy across time and states:

$$D(U\|B) \leq K.$$

In recursive form, Nature chooses current distortions and state-contingent continuation entropy.

- The uncertainty set is large and contains unstructured models.
- Nature can make q_H differ from q_L .
- The resulting optimal portfolio weights are homothetic in wealth but state-dependent.

Main result. For $\gamma > 1$, the worst-case belief is procyclical: $q_L < q_H < p$. The investor fears return persistence. For $0 < \gamma < 1$, the worst-case belief is countercyclical: $q_H < q_L < p$. The investor fears mean reversion.

4.3 Two pathwise entropy constraints: model ambiguity

Model ambiguity is represented by tighter pathwise entropy constraints. In the binomial tree, the constraints imply that continuation entropy is state-independent.

- The uncertainty set is smaller and contains structured models.
- Nature cannot allocate more future distortion to one state than another.
- The solution is equivalent to κ -ignorance in Chen and Epstein.
- Worst-case beliefs and portfolio weights become state-independent.

Interpretation. Pathwise constraints impose rectangularity. This gives dynamic consistency but rules out the state-dependent entropy allocation that captures misspecification.

4.4 Continuous-time Merton problem

The risky asset follows

$$\frac{dS_t}{S_t} = \mu dt + \sigma dB_t^B,$$

and wealth evolves as

$$dW_t = rW_t dt + \pi_t W_t ((\mu - r)dt + \sigma dB_t^B) - c_t dt.$$

Under the distorted model U , the risky asset has subjective expected return $\mu - \sigma u_t$.

The value function is homothetic:

$$V(W, E) = \frac{W^{1-\gamma}}{1-\gamma} \mathcal{V}(E).$$

Interpretation. The continuation entropy state variable captures the investor's evolving concern about misspecification, while homotheticity keeps portfolio weights independent of wealth.

4.5 Optimal distortions and effective risk aversion

For $\gamma > 1$, the optimal distortion and entropy volatility take the form

$$u^* = -\frac{(1-\gamma)\sigma}{\delta} \frac{\mathcal{V}}{\mathcal{V}_E} \pi^*, \quad g^* = -(1-\gamma)\sigma \frac{\mathcal{V}_E}{\mathcal{V}_{EE}} \pi^*.$$

The optimal risky share can be written as

$$\pi^* = \frac{\mu - r}{\sigma^2 \gamma^{\text{eff}}(E)},$$

where

$$\gamma^{\text{eff}}(E) = \gamma - (1 - \gamma) \frac{\mathcal{V}}{\delta \mathcal{V}_E} + (1 - \gamma) \frac{\mathcal{V}_E^2}{\mathcal{V} \mathcal{V}_{EE}}.$$

- The usual CRRA coefficient γ is only the first term.
- The second term captures the model ambiguity component.
- The third term captures the incremental misspecification component from state-dependent entropy.
- Effective risk aversion rises with model uncertainty, so risky investment falls below the Merton share.

4.6 Log utility

For log utility, the value function is additive in wealth and entropy:

$$V(W, E) = \log W + \mathcal{V}(E).$$

This implies $V_{WE} = 0$, so Nature has no reason to choose a nonzero entropy volatility g^* .

- The log investor is myopic.
- The log investor does not use return dynamics for hedging or speculation.
- Misspecification concerns reduce only to ambiguity concerns.

Interpretation. The log case is tractable but economically special. It hides the main distinction the paper studies.

4.7 Relation to Hansen-Sargent multiplier preferences

The paper connects the constraint approach to the Hansen-Sargent multiplier approach. The dual value function takes the form

$$\tilde{V}(W, \theta) = \inf_{E \geq 0} \{V(W, E) + E\theta\}.$$

With a fixed penalty parameter θ , the multiplier problem is generally not homothetic for CRRA preferences. In the constraint approach, the endogenous shadow price of entropy scales with wealth and remains constant along the optimal path.

Interpretation. The constraint approach is less parsimonious in state variables but more suitable for homothetic portfolio choice with general CRRA utility.

4.8 Mean-reverting Sharpe ratio extension

The final extension lets the Sharpe ratio X_t vary over time:

$$\frac{dS_t}{S_t} = \mu(X_t)dt + \sigma dB_t^B, \quad \mu(X) = r + \sigma X,$$

$$dX_t = \lambda(\bar{X} - X_t)dt + \sigma_X \left(\rho dB_t^B + \sqrt{1 - \rho^2} dW_t^B \right).$$

Alternative models can distort both the return shock and the independent Sharpe-ratio shock:

$$dE_t = \delta E_t dt - \frac{\delta}{2}(u_t^2 + v_t^2)dt + g_t dB_t^U + h_t dW_t^U.$$

- u_t distorts the expected excess return.
- v_t distorts the Sharpe-ratio drift.
- g_t and h_t make continuation entropy state-dependent with respect to return shocks and Sharpe-ratio shocks.
- Model misspecification remains tractable even when markets are incomplete.

Interpretation. Misspecification makes market timing less aggressive because Nature can both shrink current investment opportunities and make their dynamics less useful.

5 Identification and interpretation

5.1 What the paper can identify

The paper is theoretical and quantitative rather than an empirical causal design. It identifies mechanisms within a dynamic portfolio-choice model: how the investor's fear of model uncertainty changes worst-case beliefs, portfolio weights, consumption, and exit behavior.

Interpretation. The key object is not an estimated treatment effect, but a mapping from preferences and entropy constraints to worst-case beliefs and optimal policies.

5.2 Why risk-averse investors fear persistence

A risk-averse long-horizon investor with $\gamma > 1$ wants intertemporal hedging. In Merton-style logic, the investor benefits if bad current returns are followed by better future investment opportunities. The worst-case model removes this hedge by making expected returns move with returns.

- After a high return, subjective expected returns are higher.
- After a low return, subjective expected returns are lower.
- The investor behaves as if recent returns scar or reinforce beliefs.

Interpretation. The model endogenously generates extrapolative expectations for a risk-averse investor.

5.3 Why risk-tolerant investors fear mean reversion

A risk-tolerant investor with $0 < \gamma < 1$ values intertemporal speculation. This investor likes assets that pay off when future opportunities improve. The worst-case model therefore creates mean reversion, because it prevents profitable intertemporal speculation.

Interpretation. The sign of nonmyopic demand flips around log utility, so the worst-case dynamics also flip.

5.4 Why log utility is special

The log investor's portfolio is myopic. Since the log investor does not try to hedge or speculate on future opportunities, serial dependence in returns has no value for Nature to distort.

Interpretation. Log utility does not show the effect of model misspecification. It only shows pessimism about current expected returns.

5.5 Nonparticipation and belief scarring

The entropy state variable can move toward its upper boundary after bad return experiences. At the upper boundary, the worst-case subjective equity premium can fall to zero, so the investor exits the stock market.

- For patient and moderately risk-averse investors, bad experiences create persistent pessimism but not necessarily exit.
- For sufficiently impatient and risk-averse investors, a string of bad returns can generate permanent nonparticipation.
- Positive return histories can reduce entropy and lead the investor toward expected-utility behavior.

Interpretation. The model provides an uncertainty-aversion explanation for experience effects without relying on learning.

5.6 Model ambiguity versus model misspecification in practice

Model ambiguity reduces portfolio demand by lowering the current subjective equity premium. Model misspecification does more: it changes the dynamics of future beliefs and investment opportunities.

Interpretation. Misspecification nests ambiguity and adds state-dependent entropy allocation. This is why its quantitative impact is larger.

6 Tables and figures

6.1 Figure 1: Model misspecification versus model ambiguity on binomial trees

Read:

- The baseline tree has IID returns with the same high-return probability p at every node.
- Model misspecification allows q_0 , q_H , and q_L to differ, so return probabilities can depend on past returns.
- Model ambiguity uses a common distorted probability q , preserving the IID structure.

Economic message: This figure is the cleanest visual explanation of the paper. Misspecification lets Nature distort dynamics; ambiguity distorts probabilities while preserving structure.

6.2 Figure 2: IID Merton policy functions and distributions

Read:

- Panel A shows that the optimal risky share falls sharply as continuation entropy rises.
- Panel B shows that the consumption-wealth ratio also falls for the risk-averse investor, reflecting income and precautionary-savings effects.
- Panel C shows that the worst-case belief distortion rises with entropy.
- Panel D shows $g^* < 0$, meaning continuation entropy is countercyclical for $\gamma = 3$.
- Panels F-H compare simulated distributions of portfolio weights, consumption-wealth ratios, and consumption rates.

Economic message: Model misspecification lowers risky investment and consumption more than model ambiguity because it creates state-dependent pessimism about future returns.

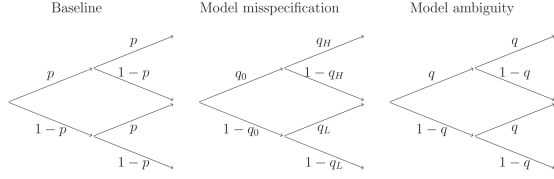


Figure 1: Model misspecification versus model ambiguity on binomial trees.

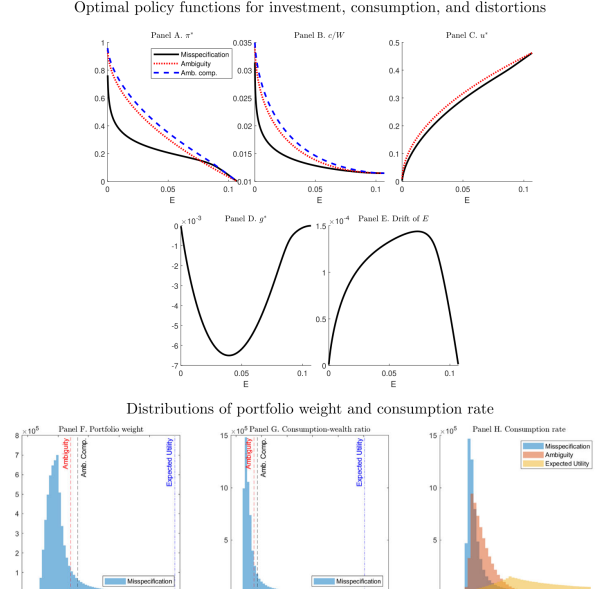


Figure 2: Optimal policy functions and distributions of optimal strategies.

6.3 Figures 3 and 4: Simulated entropy, belief scarring, and exit

Read Figure 3:

- The histograms show how continuation entropy evolves over 10-year and 50-year simulations.
- Investors who experience a very bad first year remain more pessimistic even after many years.
- Some paths eventually reach low entropy after good experiences, while high entropy is more persistent.

Read Figure 4:

- With higher risk aversion and impatience, entropy can reach the upper boundary.
- At this boundary, the investor exits the stock market.
- Negative early experiences greatly increase the probability of exit.

Economic message: Misspecification concerns generate dynamic belief scarring and can produce endogenous nonparticipation without adding participation costs or learning frictions.

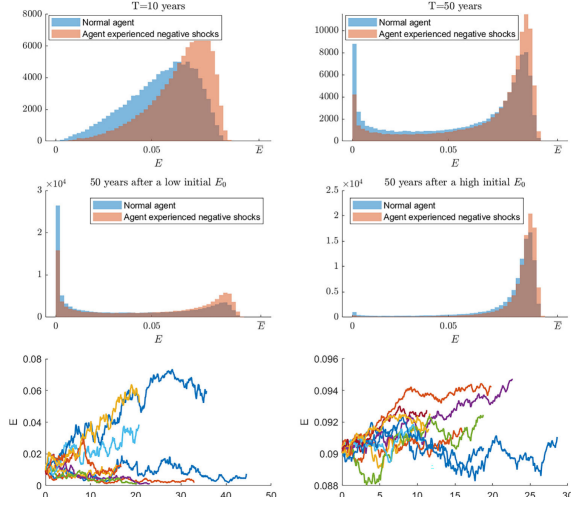


Figure 3: Histograms and sample paths of simulated entropy.

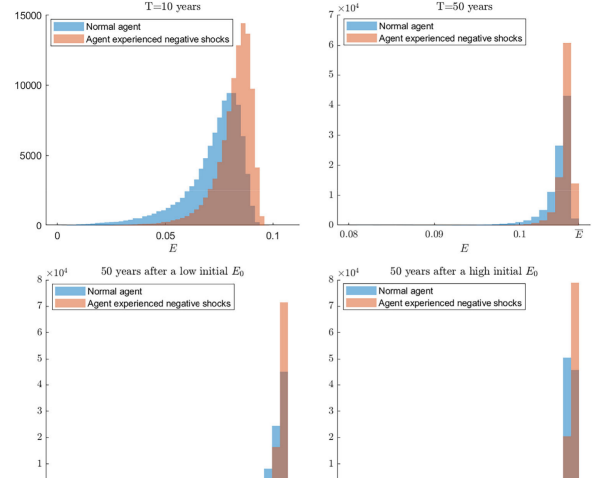


Figure 4: Endogenous exit among impatient and risk-averse investors.

6.4 Figures 5 and 6: Mean-reverting Sharpe ratio and structured versus unstructured models

Read Figure 5:

- The state variable is the conditional Sharpe ratio X , and policies are plotted for several values of entropy E .
- Model misspecification flattens the portfolio rule, reducing market timing.
- The exit boundary depends on the Sharpe ratio and entropy.
- Distortions u^* , v^* , g^* , and h^* show how Nature worsens both current opportunities and future opportunity dynamics.

Read Figure 6:

- The figure compares several structured ambiguity models with the unstructured misspecification model.
- κ -ignorance, linearly constrained models, and precommitted linear models each impose different restrictions on Nature.
- Unstructured misspecification produces lower consumption and less market timing than structured ambiguity.

Economic message: With stochastic investment opportunities, the difference between ambiguity and misspecification becomes larger because misspecification changes the dynamics of the Sharpe ratio itself.

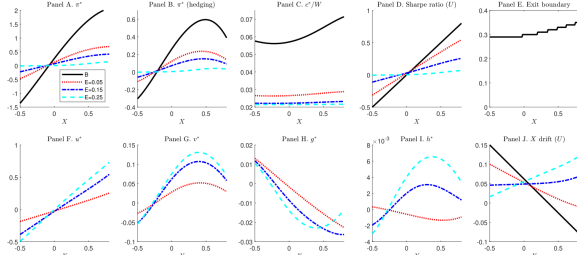


Figure 5: Model misspecification with a mean-reverting Sharpe ratio.

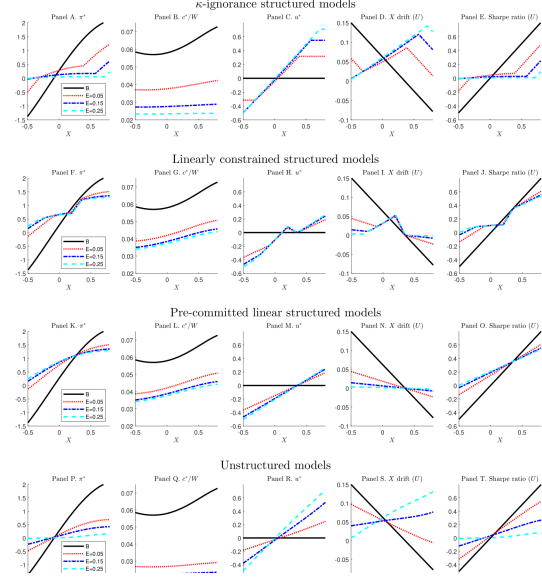


Figure 6: Model ambiguity versus model misspecification.

6.5 Figures 7 and A1: Distribution comparison and the $\gamma < 1$ case

Read Figure 7:

- Expected utility produces much more dispersed portfolio weights and consumption rates.
- κ -ignorance ambiguity reduces risk-taking substantially.
- Model misspecification further reduces portfolio weights and consumption rates.

Read Figure A1:

- For $\gamma = 0.8$, the investor is risk-tolerant rather than risk-averse.
- The entropy volatility g^* becomes positive, so the worst-case dynamics imply mean reversion rather than persistence.
- The difference between ambiguity and misspecification is smaller because $\gamma = 0.8$ is close to log utility.

Economic message: Model misspecification has quantitatively important effects, but the direction of worst-case dynamics depends on whether the investor is more or less risk-averse than log.

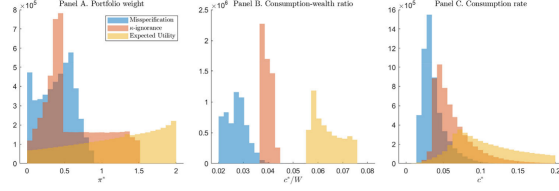
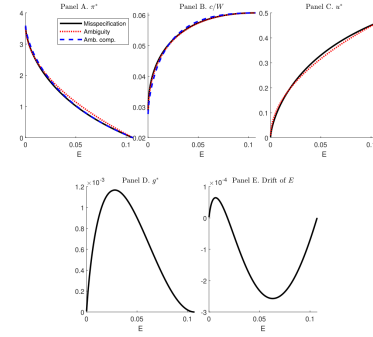


Figure 7: Distribution comparison.

Optimal policy functions for investment, consumption, and distortions



Distributions of portfolio weight and consumption rate

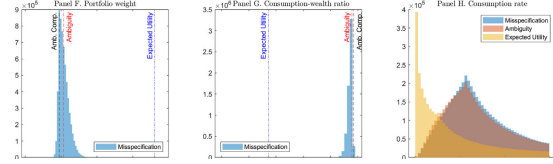


Figure A1: Optimal policy functions for the $\gamma < 1$ case.

6.6 Figures A2 and B1: Entropy for $\gamma < 1$, correlation, and risk aversion

Read Figure A2:

- For the $\gamma < 1$ case, entropy distributions are more stationary.
- Belief scarring after negative early shocks is weaker and largely fades over long horizons.
- This contrasts with the more persistent scarring in the $\gamma > 1$ case.

Read Figure B1:

- The figure studies model misspecification under different correlations and risk-aversion levels.
- When return and Sharpe-ratio shocks are uncorrelated, intertemporal hedging is absent and exit boundaries are lower.
- For $\gamma = 0.8$, the signs of the entropy responses flip relative to the $\gamma > 1$ case.

Economic message: The strength and direction of misspecification effects depend on risk aversion and on whether the risky asset can hedge changes in investment opportunities.

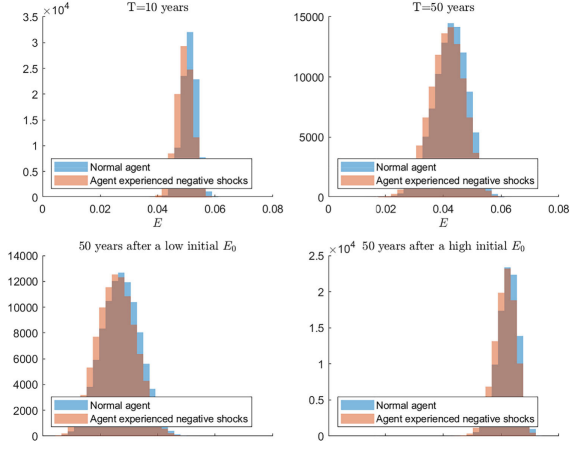


Figure A2: Histograms of simulated entropy for the $\gamma < 1$ case.

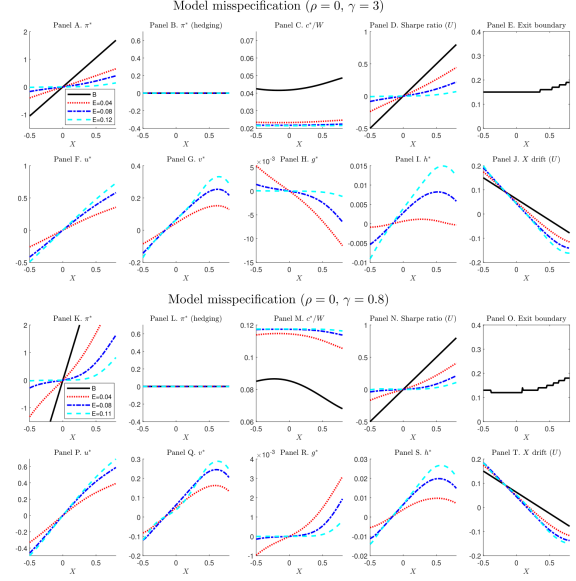


Figure B1: Model misspecification, impact of correlation, and risk aversion.

6.7 Figure C1: Ambiguity component in the misspecification solution

Read:

- Figure C1 freezes entropy at its initial value and isolates the effect of the distortion terms $u^*(E_0, X_t)$ and $v^*(E_0, X_t)$.
- The comparison separates the ambiguity component from the additional state-dependent entropy effect.
- The figure shows that state-dependent entropy explains a sizable part of the reduction in portfolio weights and consumption-wealth ratios.

Economic message: Misspecification matters not only because Nature chooses adverse distortions, but because Nature chooses how future distortion capacity moves with shocks.

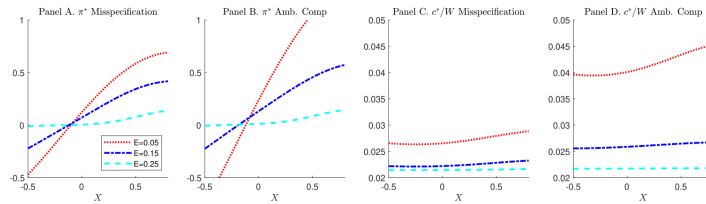


Figure C1: Ambiguity component.

7 Short summary

- The paper studies robust dynamic portfolio choice under model ambiguity and model misspecification.
- Model ambiguity means distorted beliefs within a structured model class; model misspecification allows broader unstructured alternatives.
- The central modeling device is continuation entropy, an endogenous state variable that tracks remaining distortion capacity.
- In the IID binomial tree, model misspecification lets Nature choose state-dependent probabilities q_H and q_L ; model ambiguity does not.
- For $\gamma > 1$, the worst-case model has return persistence. This removes intertemporal hedging opportunities.
- For $0 < \gamma < 1$, the worst-case model has mean reversion. This removes intertemporal speculation opportunities.
- The log investor is myopic, so misspecification collapses to ambiguity.
- In the continuous-time Merton model, effective risk aversion rises because of both ambiguity and misspecification components.
- Model misspecification reduces risky investment and consumption more than model ambiguity.
- Simulated entropy dynamics generate belief scarring: bad early stock returns can create persistent pessimism.
- For sufficiently impatient and risk-averse investors, a sequence of bad returns can push entropy to the exit boundary and generate nonparticipation.
- With a mean-reverting Sharpe ratio, misspecification reduces market timing and can distort both the level and dynamics of investment opportunities.
- The paper interprets Maenhout-style homothetic robustness as a model ambiguity case and extends the method to full misspecification.

8 Conclusion

8.1 Core conclusion

Maenhout, Xing, and Balter show that model misspecification is economically different from model ambiguity in dynamic portfolio choice. Ambiguity changes the worst-case current model within a structured set; misspecification changes the worst-case dynamics of future investment opportunities.

8.2 Economic intuition

A long-horizon investor cares about how future opportunities move with current returns. For a risk-averse investor, return persistence is bad because it prevents intertemporal hedging. For a risk-tolerant investor, mean reversion is bad because it prevents intertemporal speculation. Nature chooses exactly the dynamics that hurt the investor most.

8.3 Incremental contribution revisited

The paper’s main contribution is to make misspecification tractable without giving up homotheticity. The continuation-entropy state variable allows Nature to allocate distortion over time and states, which generates rich belief dynamics while keeping portfolio policies scale invariant.

8.4 Implications

The results imply that robust portfolio advice should distinguish between ambiguity about parameters and misspecification of dynamics. The model also provides a mechanism for extrapolative beliefs, experience effects, and stock-market nonparticipation among pessimistic investors.

8.5 Limitations

The results depend on the model structure, CRRA preferences, and entropy-based uncertainty sets. The quantitative examples use calibrated parameters rather than direct estimation. The model explains belief scarring and nonparticipation through worst-case beliefs, but actual household portfolio decisions may also reflect learning, fixed costs, constraints, and institutional factors.

8.6 Takeaway

The enduring takeaway is that model misspecification is a dynamic force. Investors do not merely fear that expected returns are lower today; they fear that the model’s entire law of motion is wrong. This changes hedging, speculation, consumption, and participation in risky assets.